

MODULE 2: PLATFORM ECONOMICS & BIG DATA

Interactive 3-Hour Rhythm

Session 2.2: Asymmetric Pricing & Subsidies

Subsidy side vs. money side economics, price elasticity, and Google Search advertising.

PART 1 (LECTURE)

60 Minutes

PART 2 (LAB PRACTICE)

60 Minutes

PART 3 (EXECUTIVE MEMO)

60 Minutes

STANDARD

Audited SEC Filings

Learning Objectives & Executive Overview

Clear conceptual grounding for undergraduate and graduate business students.

CORE COMPETENCIES

Goals

What You Will Master Today

- **Theoretical Mastery:** Evaluate seminal literature and analytical frameworks governing platform competition.
- **Empirical Rigor:** Reconcile management strategic claims with legally audited statutory filings.
- **Data Modeling:** Execute spreadsheet calculations with explicit coordinates and formula logic.
- **Executive Formulation:** Deliver structured board-level recommendations resolving complex trade-offs.

ASYMMETRIC PRICING RULE

Vocabulary

Key Topical Foundations

- **Parker & Van Alstyne:** Cross-subsidization in two-sided product design.
- **Pricing Rule:** Subsidize price-sensitive side with strong externalities.
- **Money Side:** Charge side capturing commercial surplus from interaction.
- **Micro-Examples:** Job boards (free seekers), Acrobat Reader, event venues.

Theoretical Foundations: Asymmetric Pricing Rule

Rigorous analytical frameworks and literature benchmarks.

THEORETICAL ARCHITECTURE

Literature

Seminal Framework Insights

- **Parker & Van Alstyne:** Cross-subsidization in two-sided product design.
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STRATEGIC APPLICATION

Analysis

Business Model Mechanics

- **Free Consumer Utility:** Search, Maps, Android provided at zero money cost.
- **Intent Extraction:** Aggregating commercial intent queries globally.
- **Enterprise Auction:** Auctioning targeted placements via Google Ads.
- **Revenue Dominance:** Ad revenue generated over 75% of Alphabet total.

STRATEGIC TAKEAWAY

Subsidy side vs. money side economics, price elasticity, and Google Search advertising.

Enterprise Case Dissection: Google Search Model

Historical case evidence and multi-year corporate trajectory analysis.

CORPORATE DYNAMICS

Timeline

Market Conditions & Strategic Moves

- **Free Consumer Utility:** Search, Maps, Android provided at zero money cost.
- **Intent Extraction:** Aggregating commercial intent queries globally.
- **Enterprise Auction:** Auctioning targeted placements via Google Ads.
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ANALYTICAL FOCUS

Evidence

Operational & Financial Findings

- **Alphabet 10-K Data:** Search, YouTube, Network ads, and Cloud revenue.
- **Ad Revenue Total:** Summing core advertising revenue (=SUM(B2:D2)).
- **Ad Concentration:** Ad Revenue / Total Revenue (=H2/G2*100).
- **Financial Reality:** 2023 ad revenue was \$237.8B out of \$307.4B (77.4%).

Lab Dataset & Empirical Metrics: Applied Excel Lab

Reconciling corporate filings with explicit spreadsheet formulas.

SPREADSHEET MODELING

Lab Spec

Data Metrics & Formulas

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MANAGEMENT DILEMMA

Decision Lab

Strategic Tension

- **Concentration Vulnerability:** AI chat disruption & antitrust search remedies.
- **Revenue Offset Math:** Modeling a 10% search ad decline in spreadsheet.
- **Consumer Subscriptions:** Feasibility of charging for premium AI search features.
- **Strategic Memo:** Defending free access while diversifying platform profit.

Executive Problem Solving: Executive Lab

Framing complex management dilemmas and evaluating corporate trade-offs.

EXECUTIVE CHALLENGE

Dilemma

Corporate Strategic Tension

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EVALUATION RUBRIC

Requirements

Board-Level Deliverable

- **Actionable Recommendation:** Formulate a concrete 3-bullet decision brief for senior leadership.
- **Quantitative Defense:** Support recommendations using audited financial metrics from the lab spreadsheet.
- **Risk Mitigation:** Address channel friction, partner resistance, or competitive retaliation.
- **Implementation Timeline:** Define immediate 90-day execution milestones.

Lab Transition: 120-Minute Hands-On Execution

Transitioning from lecture observation to hands-on data manipulation and strategic decision-making.

EXERCISE 1: FOUNDATIONAL PRACTICE

60 Min Lab

Applied Excel Lab

Hands-on spreadsheet protocol using audited historical numbers:

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EXERCISE 2+: ADVANCED PROBLEM SOLVING

60 Min Memo

Executive Lab

Applied scenario analysis, trade-off evaluation, and executive brief:

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